

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458745.778 +/- 0.0059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.054 +/- 0.047
Transit depth (Rp/Rs)^2	0.29 +/- 0.5 [%]
Orbital Inclination (inc)	81.68 +/- 2.87
Airmass coefficient 1 (a1)	3.31 +/- 1.28
Airmass coefficient 2 (a2)	-0.93 +/- 0.24
Scatter in the residuals of the lightcurve fit is	33.82 %
Best Comparison Star	None
Optimal Aperture	0.99
Optimal Annulus	6.0
Transit Duration (day)	0.022 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.054 ± 0.047 vs 0.1326 (deviation 0.92σ)
Duration fit vs published	0.022 d vs 0.0737 d (deviation 0.95σ)
Mid-time O–C	6.2 min
Depth SNR	0.6
Residual scatter	33.82 %

Light curve

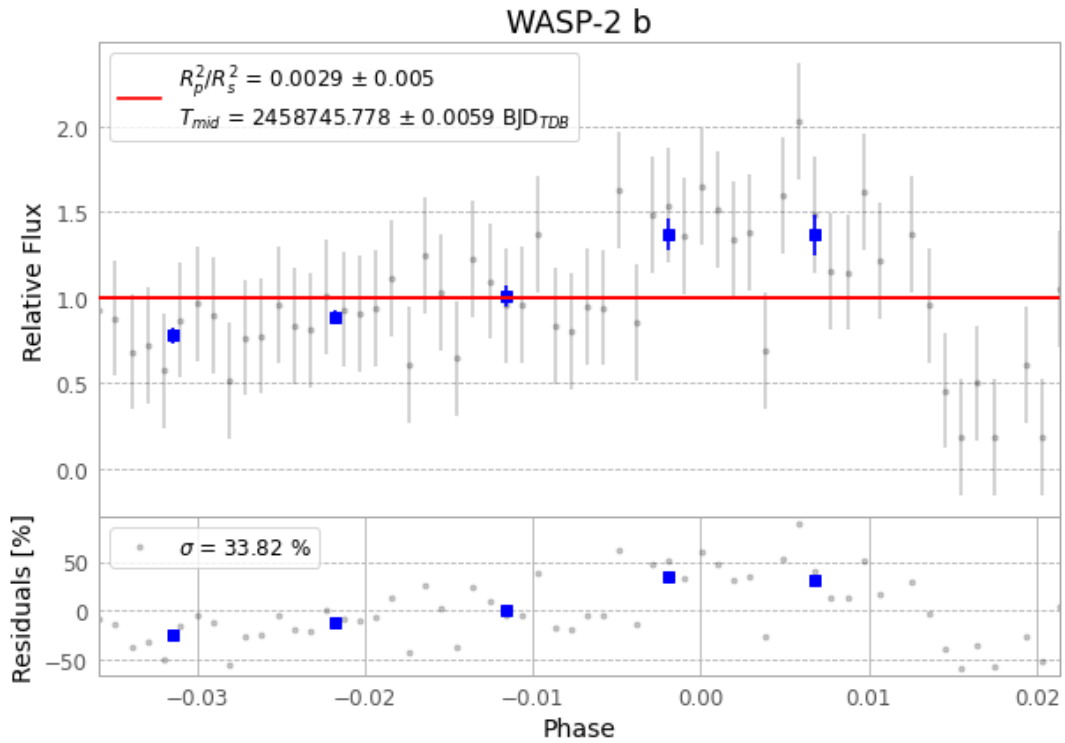


Figure 1 — FinalLightCurve_WASP-2 b_2019-09-18.png (SHA-256 5074e3ba02d6a8a9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [129, 300], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 746c9c0497cdb2c0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-2190919044212.FITS → WASP-2190919082712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-190919.log (90544ba521c9...), FinalLightCurve_WASP-2 b_2019-09-18.png (5074e3ba02d6...), FinalLightCurve_WASP-2 b_2019-09-18.csv (c22df6deb5bf...)

Compute resources

Wall time 155.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 01:16Z.